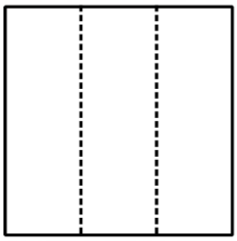
1. A square is cut into three rectangles along two lines parallel to each other as shown in the figure. If the perimeter of each rectangle is equal $\mathbf{24}$, then the area of the original square is:

A) 36 **B) 42** **C) 81** **D) 56**



2. A large rectangle is divided into four rectangles by two sections parallel to its sides. The area of three rectangles is shown as in the figure. What is the area of the fourth rectangle?

A) 15 **B) 36** **C) 102** **D) 28**

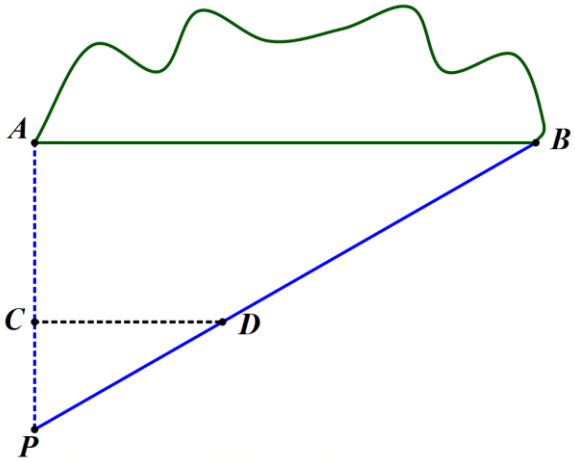
6	**14**
?	**35**

3. The sum of two numbers is equal to $\mathbf{15}$ and the sum of its reciprocal is equal to $\frac{\mathbf{3}}{\mathbf{10}}$. The two numbers are:

A) $\{8,7\}$ **B)** $\{5,10\}$ **C)** $\{4,11\}$ **D)** $\{6,9\}$

4. To find the length of the foothills, two identical triangles are constructed as shown in the figure, where PA = 180 m, CD = 150 m and PC = 50m. How long is this hill?

A) 540 m **B) 620 m** **C) 390 m** **D) 250 m**

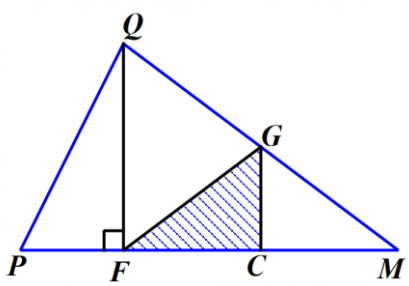


5. If $20 < p + q < 30$ and $p^2 + q^2 < 2r^2$ where **p, q** and **r** is prime number. Thus, $p + q + r$ is equal to:

A) 37 **B)** 33 **C)** 30 **D)** 28

6. Given figure below shown $\frac{FP}{FM} = \frac{1}{6}, [CG] // [QF], G$ is the center of $[MQ]$ and the area of a triangle PQM is $100\,m^2$. Find the area of the striped part (in m^2).

A) $15\,m^2$ **B)** $\frac{115}{7}\,m^2$ **C)** $\frac{150}{7}\,m^2$ **D)** $\frac{300}{7}\,m^2$

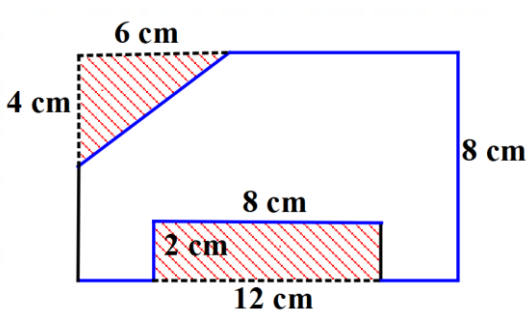


7. **ABCD** is given as a square and **AEF** is the equilateral triangle inscribed in **ABCD** square. Find the area of the square **ABCD** knowing that the area of the triangle **AEF** is equal to $\sqrt{3}$.

A) 2 **B)** $2+\sqrt{3}$ **C)** 3 **D)** $3+\sqrt{3}$

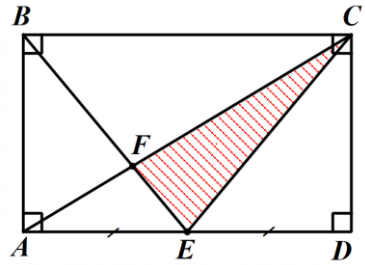
8. As shown in the figure below, the area without stripes part is equal to:

A) $128\,cm^2$ **B)** $36\,cm^2$ **C)** $68\,cm^2$ **D)** $142\,cm^2$



9. In the figure, the area of the rectangle **ABCD** is $0\ 2\ cm^2$. Find the area of the striped part.

A) $4\,cm^2$ **B)** $2\,cm^2$ **C)** $5\,cm^2$ **D)** $6\,cm^2$

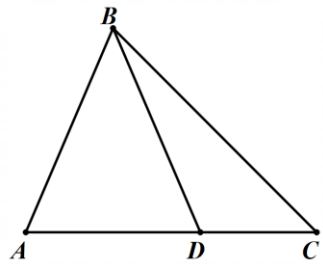


10. In the **ABCD** parallelogram, point **M** is located on the **CD**, where **3MD = 2MC**, and center **N** is located on **BM**. Thus, $\frac{The\ area\ of \Delta BCM}{The\ area\ of \Delta AND}$ is equal to:

A) $\frac{5}{6}$ **B)** $\frac{6}{7}$ **C)** $\frac{7}{8}$ **D)** $\frac{9}{10}$

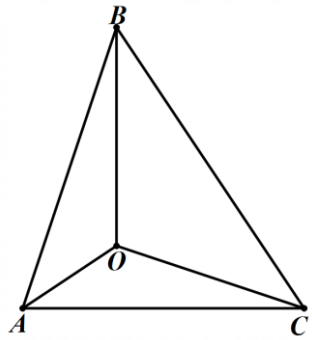
11. In the figure $AB = 5, BD = 5, BC = 7$ and $AC = 9$. Find $\frac{AD}{DC}$.

A) $\frac{17}{8}$ **B)** $\frac{17}{9}$ **C)** $\frac{18}{7}$ **D)** $\frac{19}{8}$



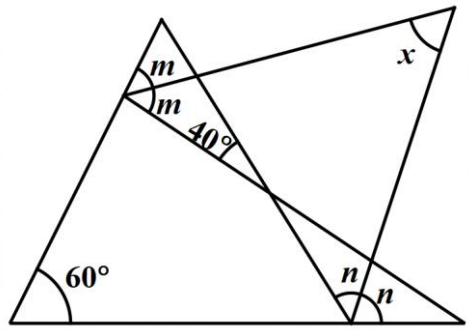
12. Inside the triangle ABC there are 36 perimeters with the point O and the length of $[OA], [OB]$ and $[OC]$ equal to $x, 2x$ and $3x$ respectively. Find the smallest value of x.

A) 3 **B)** -6 **C)** 4 **D)** 47



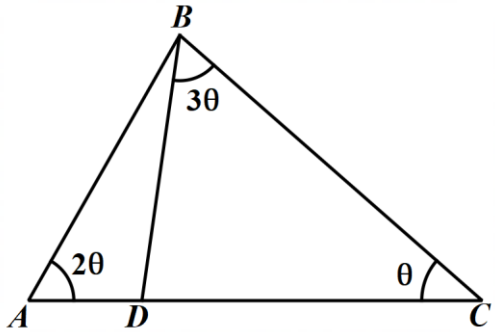
13. Find x from the figure below.

A) 40° **B) 50°** **C) 60°** **D) 45°**



14. As figure shown below $AB = 6\text{cm}, BD = 4\text{cm}$. Find DC.

A) 9 **B) 10** **C) 8** **D) 6**



15. **A, B, C, D** and **E** are 5 consecutive points on a straight line where **AC+BD+CE** = a and **BD**= b. Find **AE**.

A) $\frac{a+b}{2}$ **B)** $\frac{a-b}{2}$ **C)** $\frac{a+b}{3}$ **D)** $a-b$

16. On a straight line, observe the consecutive points **A, B,** and **C**. If **M** is the midpoint of $[AC]$, then calculate **MB**.

A) 12 **B) 9** **C) 6** **D) 3**

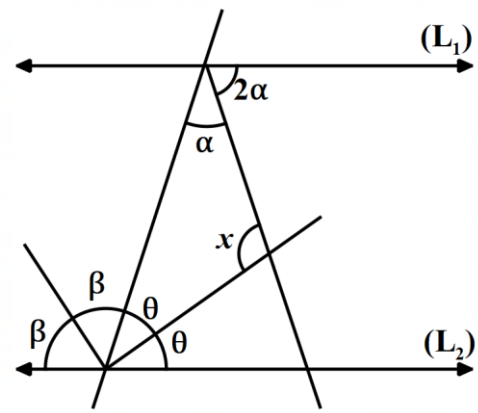
17. **A, B, C** and **D** are consecutive points on the same line where $CD = AB - 2$ and $BC = AB + 2$.

Calculates the smallest integer value of AD.

A) 7 **B) 8** **C) 6** **D) 9**

18. Base on the figure below $2\alpha - \beta > 38°$. Calculates the smallest integer of x.

A) 112° **B) 129°** **C) 119°** **D) 138°**



19. Calculate $\dfrac{\dfrac{(a+b)^2-(a-b)^2+5ab}{3ab}}{\dfrac{(a-b)^2+(b-a)^2+4ab}{a^2+b^2}}$.

A) 4 **B) $\frac{5}{3}$** **C) $\frac{1}{2}$** **D) $\frac{3}{2}$**

20. Find the sum of $3.\overline{2}$ and $5.\overline{4}$.

A) $\frac{78}{9}$ **B) $\frac{58}{3}$** **C) $\frac{58}{9}$** **D) $\frac{57}{2}$**

21. The largest five-digit numbers that can be divided by 9 and 13 is:

A) 99,972 **B) 99,918** **C) 99,945** **D) 99,964**

22. What is the multiplicative inverse of $a-\frac{1}{a}$?

A) $\frac{a}{a-1}$ B) $a+\frac{1}{a}$ C) $\frac{a}{a^2-1}$ D) $\frac{1}{a}-a$

23. Which expression is equal to the following expression?

$$-1.5+\frac{2}{5}+(-7)+2.6$$

A) $(-8.5+2.6)+\frac{2}{5}$ B) $(-5.5+2.6)+\frac{2}{5}$

C) $\left(-\frac{1}{5}+\frac{2}{5}\right)+(-9.6)$ D) $\left(-\frac{1}{5}+\frac{2}{5}\right)+(-4.4)$

24. Ms. Socheata has \$ 15.00 to pay for coffee and donuts. She bought a cup of coffee for \$ 2.59. The price of each donut is \$ 1.09. Which inequality can be used to determine the maximum number of donuts (d) that Ms. Socheata can buy?

A) $1.09d+2.59\geq 15$ B) $1.09+2.59d\leq 15$

C) $1.09+2.59d\geq 15$ D) $1.09d+2.59\leq 15$

25. Which one has the greatest value among $2^{156}, 4^{79}, 128^{23}$ and 8^{54}.

A) 2^{156} B) 4^{79} C) 128^{23} D) 8^{54}

26. The number of heads of chickens and dogs is known to be 36 and the number of legs is 100. Find the number of dogs (chickens have 2 legs and dogs have 4 legs).

A) 8 B) 34 C) 14 D) 22

27. Three devices receive signals from three different locations: the first device sends a signal every 10 minutes, the second device sends a signal every 15 minutes, and the third device sends a signal every 20 minutes. Find the shortest time that all three devices give the same signal again after it has given the same signal at the same time (in hours).

A) 2 **B) 1** **C) 3** **D) 4**

28. In Rumduol's tour, on the first day she spends $\frac{7}{24}$, on the second day she spends $\frac{1}{4}$ and on the third day she spends $\frac{1}{3}$ of her money. Find the fraction of Rumduol's remaining money to spend on the fourth day.

A) $\frac{1}{8}$ **B)** $\frac{7}{2}$ **C)** $\frac{1}{4}$ **D)** $\frac{3}{4}$

29. The cost of a Vanda lunch at a restaurant is $ 13.00. He gives the waiter 17% of the lunch price. What is the total his expenses?

A) $2.21 **B) $10.79** **C) $13.17** **D) $15.21**

30. A car can drive 350 km in 4 hours. If its speed drops by 12 km / h, how long will it take for the car to reach 450 km?

A) 8h **B) 12h** **C) 6h** **D) 4h**